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Safety Studies Using Artificial Intelligence

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Abstract

This paper presents a comprehensive exploration of how Artificial Intelligence (AI), particularly reinforcement learning, can enhance the hazard identification within process safety studies such as Hazard and Operability (HAZOP) analysis. Traditional safety assessment methods, while foundational to standards like IEC 61508, are increasingly challenged by their reliance on expert judgment, which introduces subjectivity, cognitive bias, and procedural inconsistencies. These limitations are further exacerbated by the resource-intensive nature of manual workshops and the loss of institutional knowledge due to workforce turnover. To address these challenges, the study introduces a novel AI-based approach using multi-agent reinforcement learning, where simulated agents interact within dynamic plant environments—specifically an LNG train and a boiler system—to identify and respond to hazards. The AI model, trained over 1,000 episodes, demonstrated superior hazard coverage and scenario diversity compared to manual methods, enhancing both the speed and reliability of risk assessments.

Beyond initial hazard identification, the paper explores AI's potential to validate the effectiveness of protection layers during operation by comparing real-time process data against design assumptions. This capability supports continuous safety assurance throughout the lifecycle. The study emphasizes the importance of high-quality training data, expert oversight, to ensure model reliability to the maximum extent possible. While AI introduces non-deterministic behavior that complicates validation, the integration of human-in-the-loop processes and robust qualification frameworks can mitigate these risks. Ultimately, the paper concludes that AI has the potential to augment traditional safety practices, offering a scalable, objective, and intelligent solution for modern industrial hazard identification and safety lifecycle management.

Introduction

Hazard identification is a fundamental pillar of the functional safety lifecycle as defined by IEC 61508, yet the traditional methods used to perform it—such as HAZOP and HAZID—are increasingly showing their age. These techniques, while well-established, rely heavily on expert-led workshops that are both time-consuming and resource-intensive. Their effectiveness is often shaped by subjective judgment, making them

vulnerable to cognitive biases, procedural inconsistencies, and the inevitable loss of institutional knowledge as experienced professionals retire or transition out of roles. This reliance on human expertise creates gaps in hazard recognition and risk evaluation, which are frequently cited as root causes in accident investigations. Moreover, consequence analysis in these methods often lacks sufficient depth, leading to underestimated risks and overlooked scenarios

A study by the Health and Safety Laboratory (Year 2000) identified over 40 hazard identification techniques, each suited to different stages of process design. (Reference HSL/2005/58)

IEC 61511 requires that hazard and risk assessments be conducted regardless, of the method chosen. The choice of techniques depends on factors like process complexity, available expertise, and budget. Common techniques include HAZOP, What-If Analysis, FMEA, Fault Tree and Event Tree Analysis, and LOPA—each offering specific advantages depending on the context.

As industries seek faster and more efficient approaches, Artificial Intelligence (AI) is emerging as a promising tool. Its ability to process large and diverse volumes of data, from differing sources, in a shorter time can enhance the speed and accuracy of hazard identification. However, its effectiveness depends on well-trained models, high-quality data, and human oversight to validate results. When trained on historical incidents and hazard data, AI can help improve the consistency and reliability of risk assessments, working alongside traditional methods rather than replacing them,

While AI can enhance safety design, human oversight remains essential due to the complexity and stringent regulatory requirements as well as uncertainty in some cases about the validity of the outcomes (unless thoroughly validated). Compliance with standards such as IEC 61511 mandates that accountability in safety lifecycle activities lies with qualified and competent professionals. Moreover, the reliability of AI-driven insights depends heavily on the quality, transparency, and traceability of the underlying data and models.

Importantly, human judgment brings essential capabilities such as critical thinking, contextual understanding, and ethical reasoning, skills that are indispensable when making safety-related decisions and interpreting AI outputs within the broader operational and regulatory framework.

The integration of Artificial Intelligence into HAZOP practices introduces a range of new challenges, particularly in the qualification and validation of identified hazards. One of the key concerns stems from the inherently non-deterministic nature of AI outputs. Unlike traditional deterministic systems, AI-based tools may yield different results when the same analysis is repeated, or they may generate inaccurate or misleading outcomes if the model training data does not adequately represent the simulated scenarios. This variability can undermine confidence in the consistency and reliability of hazard identification.

Furthermore, AI systems often operate as "black boxes," making it difficult to interpret the rationale behind their decisions or to guarantee their performance under all conditions. This lack of transparency poses a significant challenge, especially when AI is used to support safety-critical tasks such as hazard analysis. As a result, the deployment of AI in these contexts necessitates special considerations, including rigorous validation protocols and steps, traceability of decision-making processes, and continuous oversight by qualified safety professionals and experts.

These complexities can complicate the overall safety assessment process. However, with the implementation of robust model training strategies, the use of diverse and high-quality data sources, and the integration of expert human judgment, it could be possible to mitigate these risks. This underscores the importance of developing and adhering to comprehensive qualification frameworks that account for the probabilistic and adaptive behavior of AI systems, while ensuring that safety and reliability remain paramount in process design and operational decision-making. This paper focused on the use case of AI in the process of hazard identification studies as well on training the AI model

Statement of Theory and Definitions

So let us briefly analyze ML (Machine learning) method considered in this experience. In simple terms, machine Learning can be broadly divided into three types:

- **Supervised Learning**, where models are trained on labeled data to make predictions (e.g., classification and regression tasks)
- **Unsupervised Learning**, which involves discovering patterns or groupings in unlabeled data (e.g., clustering and dimensionality reduction)
- **Reinforcement Learning**, where models learn optimal actions through trial and error by interacting with an environment and receiving feedback in the form of rewards or penalties.

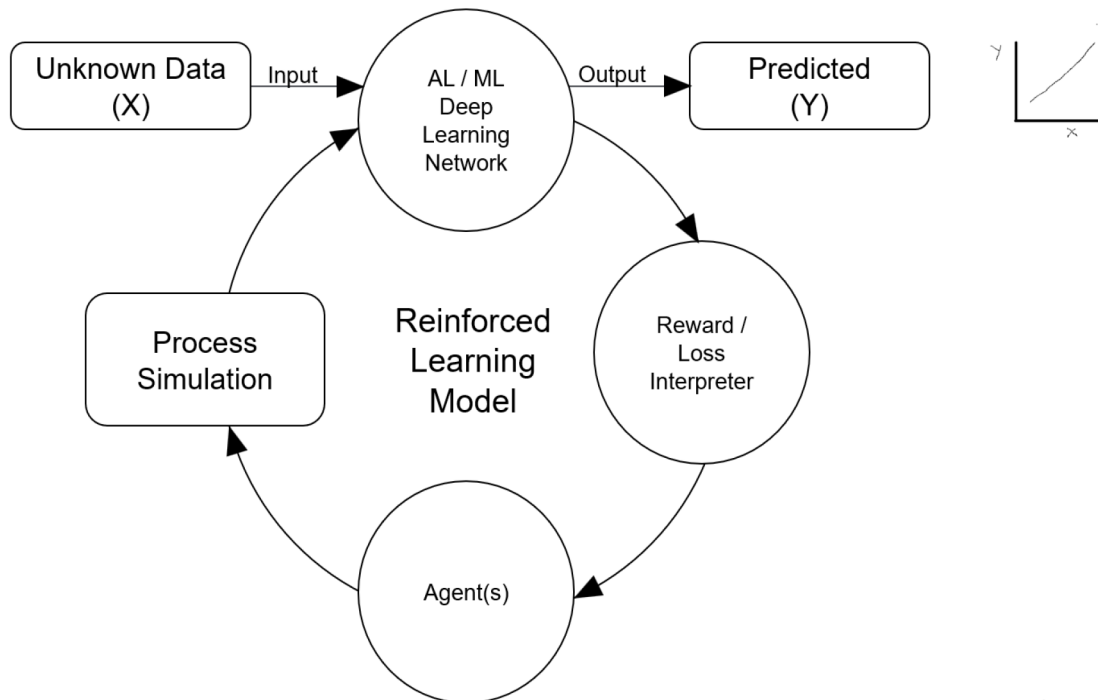


Figure 1—Reinforced learning model

Each type serves different purposes and is applied depending on the nature of the problem and the available data. The idea of applying AI to the HAZOP process is not entirely new; it has been a topic of discussion for several decades. This paper builds on that foundation by introducing a novel approach inspired by OpenAI's "Hide and Seek" simulation, in which AI agents learn complex, adaptive strategies through interaction and competition. This mirrors the dynamic interaction between hazards and safeguards in industrial systems.

In the simulation, "hiders" (good agents) creatively used their environment to avoid detection (barriers), while "seekers" (bad agents) learned to overcome these defenses, leading to emergent behaviors not explicitly programmed (threats).

When applied to the HAZOP framework, hazards act as bad agents trying to exploit system vulnerabilities, while safeguards are good agents designed to detect, prevent or mitigate those risks. This agent-based perspective reflects the dynamic interaction between hazards and risk reduction methods.

By simulating these interactions in a controlled environment, AI can learn how hazards evolve and how safety systems respond, uncovering hidden vulnerabilities and testing the effectiveness of protective measures.

This multi-agent reinforcement learning approach enables the AI to explore a wide range of "what-if" scenarios, making it a powerful tool for dynamic, automated hazard identification and risk assessment. Reinforced learning involves training agents through iterative reward-based feedback mechanisms.

In this study, two agents operate within a simulated LNG process: the Bad Agent introduces faults, while the Good Agent detects hazards. Terms such as overpressure, overtemperature and transmitter failure (the bad agents) are just some of the deviations defined within the context of process hazard analysis.

The model could generate a summary of the hazard scenarios, recommended safeguards as well recommended operating, alarming and trip parameters.

Description and Application of Equipment and Processes

Training an AI model to perform a HAZOP process, several complementary approaches can be used to ensure accuracy and reliability.

First, historical HAZOP studies and incident reports can be collected and annotated to train the AI model in recognizing hazards, causes, and safeguards based on past patterns. This approach enables the model to learn from established patterns and expert reasoning, thereby enhancing its ability to replicate and support structured hazard identification.

Second, dynamic plant simulations using Operator Training Simulators (OTS) allow the AI to observe realistic operational scenarios, including fault conditions and abnormal events, providing valuable time-sequenced data to understand risk behavior, operator responses and the potential progression of hazardous conditions under varying conditions.

Third, AI can be trained to interpret process flow diagrams and P&IDs, automatically identifying nodes and predicting potential deviations.

A human-in-the-loop approach is also essential, where safety experts validate and refine AI-generated assessments to improve model accuracy over time.

Additionally, building ontology-based knowledge graphs helps the AI understand complex interrelationships between process elements, hazards and risk reduction methods.

Finally, integrating near real-time process data enables the AI to detect anomalies and potentially predict risks proactively (move towards generative AI)

Plant simulation, especially Operator Training Simulators (OTS), is considered an effective way to train AI models for hazard analysis, because it provides realistic, dynamic data that closely mirrors plant operations, including fault / hazard insertion scenarios and operator responses. This allows the AI to learn complex cause-effect relationships, system behaviors, and human factors in a safe, controlled environment.

Simulations offer a wide range of diverse scenarios, continuous learning opportunities, and integration with detailed process models, making them well suited for developing AI systems capable of identify and assess hazards with minimal bias and high reliability.

Presentation of Data and Results

In this example, two simulation models were used 1) an LNG Train and 2) a boiler system, both using high-fidelity process simulators.

In each, the bad agent introduces faults like overpressure and over temperature, while the good agent identifies resulting hazards. Each simulation episode includes fault initiation and hazard detection, with agents receiving rewards based on performance metrics.

The plant model serves as the simulated environment, while a machine learning engine, specifically a reinforcement learning model, drives the decision-making process.

Two types of agents operate within this environment: the Bad Agent, which attempts to create process hazards by disrupting plant operations, and the Good Agent, which counters these threats by implementing mitigations such as adjusting process parameters or initiating shutdowns.

The simulation generates a report detailing identified hazards and corresponding mitigation strategies. Once validated, this output can be used to automate various stages in the process safety lifecycle, offering a dynamic and intelligent approach to traditional hazard analysis.

Initial trials with 1,000 iterations showed that the Bad Agent quickly learned to initiate hazards, while the Good Agent required 300–400 iterations to consistently detect them.

However, faster simulations without accuracy led to reduced hazard detection. AI-generated scenarios were more diverse and consistent than manual methods, improving confidence in hazard identification.

The research demonstrated the AI model's ability to simulate agent behavior under various process conditions and assess hazard responses. By analyzing combinations of agent actions, categorized as either bad (e.g., decreasing water or steam, increasing fuel or air, or taking no action) or good (primarily system shutdowns), the model effectively identified critical hazards such as low water levels, high pressure, and high temperature.

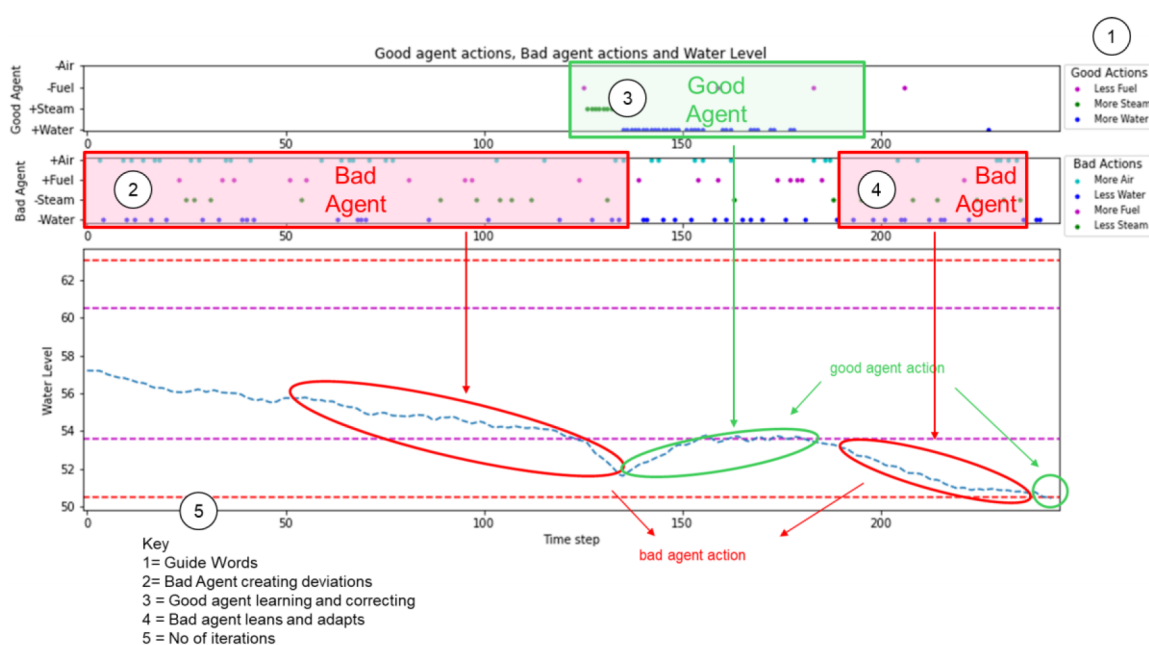


Figure 2—Agent behavior

The frequency ranking of bad actions across simulations provided insight into common failure patterns, while the comparison against predefined safety limits validated the model's responsiveness. These results confirm the potential of using AI to support real-time hazard identification and enhance decision-making process.

While a traditional HAZOP workshop for this equipment typically requires two to three days depending on its complexity, the AI model completed the simulation within minutes. This significant time reduction was made possible by leveraging an existing process simulation model of a similar unit, which served as a prerequisite for the AI analysis

Model results verification techniques

Building confidence in the outputs of the AI-driven HAZOP model requires a structured and ongoing validation process. This involves periodically comparing the model's hazard scenarios with actual HAZOP workshop reports and benchmark datasets to ensure alignment with established safety practices. A practical approach includes testing the model against previously completed studies to assess its accuracy and

reliability, in short, would the model generate the same outcomes or not, and why, and if a GAP was identified, then a re-learning process is required.

While this validation doesn't need to occur with every use, it should be repeated until the model demonstrates consistent and steady performance with minimum nuisance. In many cases, the AI model can also serve as a valuable tool to support traditional HAZOP workshops by generating preliminary hazard scenarios based on high-quality input data, such as historical incidents and process documentation in conjunction with a process simulation software for a similar units / process under study if possible. These insights can streamline discussions and possibly reduce preparation time, though this application was beyond the scope of the current study and remains a promising area for future exploration.

To support this verification process, a structured validation framework can be developed, including a comprehensive checklist focused on data quality, model transparency, and traceability. Ensuring data quality helps prevent misleading outputs, while transparency makes the model's decision-making process interpretable and auditable—critical in safety-critical environments. Traceability allows each hazard scenario to be linked back to its source data, supporting accountability and reproducibility. A detailed flowchart can also be created to guide the validation process step-by-step, ensuring systematic evaluation. These measures help ensure that AI enhances hazard identification without compromising the integrity of the safety assessment process

Use cases in change management

During the operation and maintenance phase of the safety lifecycle, an AI-trained HAZOP model can play a pivotal role in supporting the Management of Change (MoC) process, particularly when revalidating hazard analyses. As operational conditions evolve—whether due to equipment upgrades, process modifications, or control strategy adjustments—the AI model can be used to simulate the impact of these changes on system safety. By analyzing real-time process data and comparing it against original HAZOP assumptions, the model can identify emerging hazards or shifts in risk profiles that may not be immediately apparent. This capability allows for proactive hazard revalidation, ensuring that safety measures remain effective and relevant. Additionally, the AI model can generate preliminary hazard scenarios for proposed changes, serving as a decision-support tool for MoC review teams. These insights can streamline workshop discussions, prioritize risk mitigation efforts, and enhance traceability in safety documentation. When integrated with digital twins and supported by expert oversight, the AI model becomes a dynamic asset in maintaining safety integrity throughout the lifecycle, reinforcing compliance with standards such as IEC 61511 and improving the overall robustness of the MoC process

Safety digital Twin

The study explored the use of AI in re-validating the effectiveness of protection layers (safeguards) during operating and maintenance stage of the safety lifecycle. By collecting real-time data from the operating process and comparing it with the original design assumptions, the AI model can identify discrepancies or gaps between the actual Risk Reduction Factor (RRF) delivered by each safeguard and the expected design values, potentially offering actionable insights in near real time.

To support this capability, the AI model must be trained using existing plant documentation, including HAZOP, Layer of Protection (LOPA), and Safety Requirement Specifications (SRS) studies, while concurrently monitoring critical parameters from the live process. This approach ensures that the design phase assumptions are continuously validated against actual operating conditions.

Integrating a safety digital twin with an AI-trained HAZOP model offers a powerful, real-time approach to hazard identification and risk management. The digital twin acts as a live, virtual replica of the physical plant, continuously reflecting operational data such as process conditions, equipment status, and control logic. By feeding this real-time data into the AI model, the system can dynamically simulate and assess

potential deviations, enabling early detection of emerging hazards. This integration allows for continuous validation of HAZOP assumptions, especially during the operation and maintenance phase, and supports proactive decision-making in response to changes.

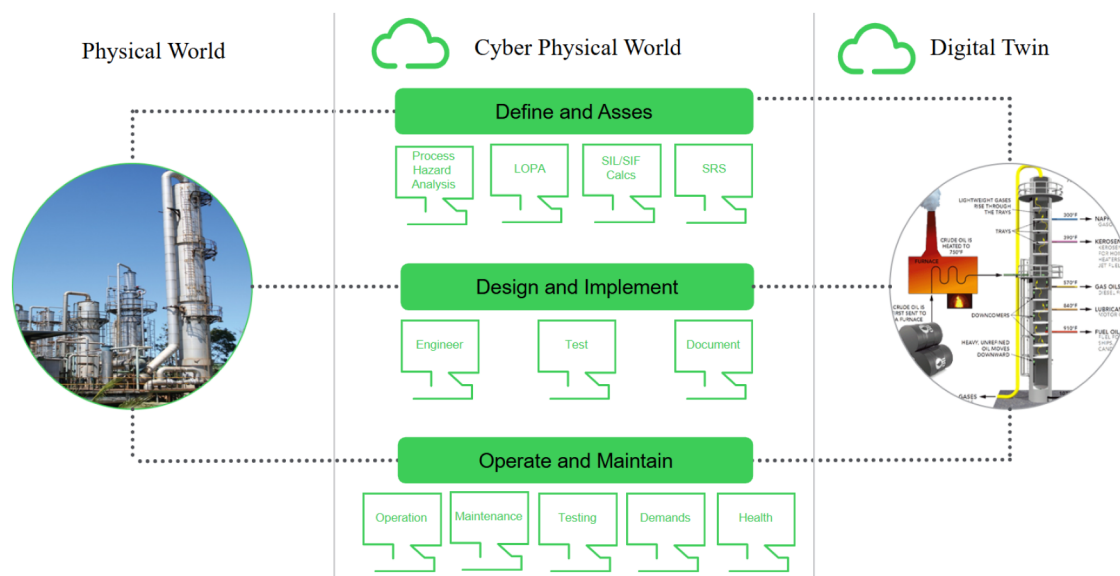


Figure 3—Safety Digital Twin

When used in conjunction with Management of Change (MoC) processes, the digital twin can simulate proposed modifications, while the AI model evaluates their safety implications—helping teams identify risks before implementation. The combined system enhances situational awareness, improves the accuracy of hazard assessments, and ensures that safety measures remain aligned with actual plant conditions. Ultimately, this synergy supports a more adaptive, predictive, and resilient safety management strategy.

Conclusions

In conclusion, this study demonstrates the transformative potential of Artificial Intelligence—particularly reinforcement learning—in enhancing hazard identification and safety lifecycle management within industrial environments. By simulating dynamic interactions between hazards and safeguards, AI models can uncover complex scenarios that traditional methods may overlook, offering improved consistency, speed, and depth in risk assessments. The integration of AI into HAZOP practices, while introducing challenges such as non-deterministic behavior and validation complexity, can be effectively managed through structured qualification frameworks, high-quality data, and expert oversight. Furthermore, the application of AI during the operation and maintenance phase, especially within the Management of Change (MoC) process, enables proactive hazard revalidation and supports continuous safety assurance. As standards like IEC 61508 and ISO/IEC TR 5469 evolve to accommodate AI technologies, the findings of this paper underscore the importance of combining intelligent systems with human judgment to build resilient, adaptive, and compliant safety strategies. Future research should continue to explore the integration of AI with digital twins, explainable AI, and generative models to further advance the field of functional safety.

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Nomenclature

PHA	Process Hazard Analysis
HAZOP	Hazard and Operability Study
HAZID	Hazard Identification
AI	Artificial Intelligence
IEC	International Electrotechnical Commission
ISO	International Organization for Standardization

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